

MAX LEONG RUI SHENG

Singapore | 86835681 | maxleong32@gmail.com
github.com/maxi-cmyk | linkedin.com/in/maxleongruiheng

EDUCATION

Singapore Management University (SMU)

Expected 2029

Bachelor of Science in Computing (Computer Science)

WORK EXPERIENCE

DSO National Laboratories — Lab Assistant

Feb – Apr 2023

- Prepared equipment and materials for laboratory experiments, ensuring readiness and safety compliance for daily research operations
- Synthesized chemical compounds in support of ongoing research projects, following established protocols and safety procedures

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++

Frontend & Product: React, Next.js, Electron, Vite

Backend & Data: FastAPI, Supabase, Redis, Celery, SQLite, WebSockets

AI & Computer Vision: MediaPipe, OpenCV, Librosa, Whisper

Hardware: ESP32, Arduino, Blynk

CERTIFICATIONS

Google AI Professional Certificate — Coursera, Jun 2026

[Verify](#)

HACKATHON ACHIEVEMENTS

it'sPEAK (Presentation Coach) — 2nd Place, SMU .Hack Enrichment Application Programme (HEAP) 2026 | [GitHub](#) | *Python, FastAPI, Next.js, MediaPipe, Librosa, Whisper, Celery, Redis, Supabase*

- Built a private presentation-practice coach that turns a short recording into evidence-based feedback on pacing, pitch variation, filler words, pause quality, eye-contact proxy, posture, and gesture range
- Implemented a baseline-comparison system letting users measure rehearsal progress against a protected personal benchmark
- Designed an asynchronous processing pipeline (Celery + Redis) to handle audio/video analysis without blocking the user-facing app

Echo — Best Freshmen Award, NTU iNTUition 2026 | [GitHub](#) | *TypeScript, Next.js, Supabase, Computer Vision*

- Developed a reminiscence-therapy companion for people with dementia, prioritising familiar memories over novel content to reduce cognitive overload
- Designed adaptive interaction logic that adjusts pacing and content during detected moments of user vulnerability

PERSONAL PROJECTS

Software

Network Sniffer | [GitHub](#) | *Python, Scapy, FastAPI, WebSockets, Next.js*

- Built a real-time packet capture and telemetry dashboard, streaming live network traffic data from a Scapy backend to a Next.js frontend over WebSockets
- Architected a FastAPI service layer to process and relay packet-level data with minimal latency

Hologram AR | [GitHub](#) | *Python, MediaPipe, Computer Vision*

- Created a real-time webcam AR system using MediaPipe hand tracking to drive interactive overlays and gesture-triggered visual effects
- Engineered spatial interaction logic mapping tracked hand landmarks to on-screen holographic elements

Hardware

Conway's Game of Life Data Factory | [GitHub](#) | *ESP32, React, Jupyter Notebook*

- Implemented Conway's Game of Life on a 32×8 LED matrix driven by an ESP32, streaming live entropy and birth/death telemetry to a React dashboard

ESP32 Asteroids | [GitHub](#) | *ESP32, C++, OLED, PWM*

- Built a breadboard-based Asteroids arcade game on an ESP32 with a 128×64 OLED display, joystick controls, and PWM-driven sound effects
- Implemented persistent high scores, adaptive difficulty, and an AI-driven attract mode

CO-CURRICULAR ACTIVITIES

SMU Legal Innovation & Technology Society — Law Associate (EXCO)

- Contribute to a cross-disciplinary CCA (School of Law) exploring the intersection of legal practice and technology
- Authored a published policy paper examining agentic AI liability and indirect prompt injection under Singapore's Computer Misuse Act (CMA) and PDPA
- Analyzed zero-click AI vulnerabilities (EchoLeak, CurXecute, BodySnatcher) to propose statutory safe harbors under PDPA § 24 and CMA amendments criminalizing adversarial prompt injection